CASE 4

Mrs. Catherine Holt — 52F · SPMS | 2-year progressive decline · Rolling frame (all indoor mobility) · Bilateral adductor spasticity · Personal care and transfers impaired · Lhermitte's sign

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

• Problems Identified
• Physical Impairments
• Bilateral lower limb spasticity — adductors predominant (UMN pattern, progressive)
• Significant bilateral lower limb weakness (rolling frame required for all indoor mobility)
• Cervical cord vulnerability — Lhermitte's sign, restricts treatment positions
• Likely pain secondary to spasticity (groin, hip, adductor origin)
• Likely bladder / bowel dysfunction (spasticity precipitant + independent impairment)
• Functional Activity Limitations
• All transfers (bed, chair, toilet) compromised by spasticity and weakness
• Personal care limited — adductor spasticity closes legs, impeding perineal access
• All indoor mobility requires rolling frame; outdoor mobility likely absent or minimal
• Dependence on carer for most ADLs

■ RED FLAGS

• Sudden or significantly worsening Lhermitte's sign
• Rapidly escalating spasticity over days or weeks
• New or worsening pressure ulcer / skin breakdown
• Autonomic dysreflexia features (if high spinal cord involvement suspected)
• Signs of deep vein thrombosis in lower limb

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Modified Ashworth Scale (MAS) — Bilateral Hip Adductors Spasticity
Test 1: Modified Ashworth Scale (MAS) — Bilateral Hip Adductors Spasticity What it measures: Resistance to passive movement — grades spasticity on a 0–4 scale (0 = no increase; 1 = slight catch; 1+ = catch with minimal resistance; 2 = marked increase but passive movement possible

Passive Range of Motion — Hip Abduction (Goniometry) Contracture Screen
Test 2: Passive Range of Motion — Hip Abduction (Goniometry) Contracture Screen What it measures: Available hip abduction range bilaterally, distinguishing flexible (dynamic) spasticity from fixed musculotendinous contracture. This distinction determines whether stretching is lik

Timed Up and Go (TUG) with Rolling Frame Functional Mobility
Test 3: Timed Up and Go (TUG) with Rolling Frame Functional Mobility What it measures: Functional mobility and fall risk using Catherine's actual mobility aid — giving an ecologically valid snapshot of her real-world transfer and walking capacity. How to perform: Seat Catherine i

■ TREATMENT PLAN

■ 1. Identify and Remove Spasticity Precipitants

■ 2. Spasticity Management — Positioning and Daily Stretching

■ 3. Transfer Training and Carer Manual Handling Education